



BriteTest

Test Reference

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This document summarizes the public Test API declared in `include/testapi.h`. It focuses on the helper-specific return conventions and the option structs used by the comparison helpers.

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Preface

This document is intended for BriteTest users and contributors who need quick access to the definitions of terms used in BriteTest or to browse through the terms.

For a list of other BriteTest documents and the BriteTest repository layout, see the Documentation Guide.

For a glossary of terms, see the Glossary Reference.

Document Version History

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1.0	1.0.0	1.0.0	2026-06-11	Initial version.	Paul Sinclair

- The **Document** column records the document's version with the format **M.u** (Major, update).
- The **Runner** column records the BriteTest Runner API version current at the time this document version was published and is defined by its `RA_RUNNER_VERSION` macro.
- The **Test** column records the BriteTest Test API version current at the time this document version was published and is defined by its `RA_TEST_VERSION` macro.
- Both Runner and Test use the version format "**M.m.p**" (Major, minor, patch).
- **M** is the same for the Document, Runner, and Test versions.

The document's update version tracks released updates to this document and does not correspond to a minor or patch version. **u** increments when document changes are published in a release without a change to **M**, and it resets to 0 when **M** is incremented.

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1. Return Conventions

1.1 Return Styles

BriteTest test helpers use three return styles:

- Structured helpers return `RA_OK`, `RA_MISMATCH`, `RA_TIMEOUT`, or an `RA_E*` error code.
- Predicate helpers return 1 for true or match, 0 for false or no match, and -1 on invalid input or runtime error.
- Process helpers that model command completion return 0 for success, 1 for timeout, and -1 for invalid input or setup failure.

1.2 Assertion Guidance

Use the helper family to decide what to check:

- File, JSON, metadata, and normalized-text comparisons should usually be checked against `RA_OK` or `RA_MISMATCH`.
- Existence, string, wildcard, and regex predicates should usually be checked against 1 or 0.
- Command-execution helpers should usually be checked against 0 or 1.

2. Comparison Option Structs

The comparison helpers use small option structs instead of mixed boolean and flag parameter lists.

2.1 `ra_path_compare_options_t`

Used by `ta_compare_paths_with_options`.

```
typedef struct {  
    int flags;  
} ra_path_compare_options_t;
```

Supported flags:

- `RA_FILECMP_IGNORE_TRAILING_WHITESPACE`
- `RA_FILECMP_IGNORE_EMPTY_LINES`
- `RA_FILECMP_IGNORE_TIMESTAMPS`
- `RA_FILECMP_IGNORE_LINE_ENDINGS`
- `RA_FILECMP_CASE_INSENSITIVE`

Default initializer:

```
#define RA_PATH_COMPARE_OPTIONS_INIT {0}
```

2.2 `ra_path_metadata_compare_options_t`

Used by `ta_compare_path_metadata`.

```
typedef struct {  
    int flags;  
} ra_path_metadata_compare_options_t;
```

Supported flags:

- `RA_STATCMP_SIZE`
- `RA_STATCMP_PERMS`
- `RA_STATCMP_MTIME`
- `RA_STATCMP_OWNER`
- `RA_STATCMP_TYPE`

Default initializer:

```
#define RA_PATH_METADATA_COMPARE_OPTIONS_INIT {0}
```

Passing `NULL` or the default initializer uses the common default metadata set: size, permissions, and type.

2.3 `ra_json_compare_options_t`

Used by `ta_compare_json_with_limit`.

```
typedef struct {  
    size_t max_bytes;
```

```
    int ignore_key_order;  
} ra_json_compare_options_t;
```

Default initializer:

```
#define RA_JSON_COMPARE_OPTIONS_INIT {0, 0}
```

Use `ignore_key_order` when semantic JSON equality matters more than source ordering.
Use `max_bytes` when the caller wants a tighter bound than the implementation default.

2.4 `ra_text_compare_options_t`

Used by `ta_compare_text_normalized`.

```
typedef struct {  
    int ignore_whitespace;  
    int ignore_line_endings;  
} ra_text_compare_options_t;
```

Default initializer:

```
#define RA_TEXT_COMPARE_OPTIONS_INIT {0, 0}
```

3. Common Usage Patterns

3.1 Compare Two Paths While Ignoring Timestamps

```
const ra_path_compare_options_t options = {
    RA_FILECMP_IGNORE_TIMESTAMPS
};
```

```
RA_ASSERT(ta_compare_paths_with_options(expected_path, actual_path, &options) == RA_OK,
```

3.2 Compare Path Metadata Including Modification Time

```
const ra_path_metadata_compare_options_t options = {
    RA_STATCMP_SIZE | RA_STATCMP_PERMS | RA_STATCMP_TYPE | RA_STATCMP_MTIME
};
```

```
RA_ASSERT(ta_compare_path_metadata(left_path, right_path, &options) == RA_OK, 0);
```

3.3 Compare JSON While Ignoring Object Key Order

```
const ra_json_compare_options_t options = {
    0,
    1
};
```

```
RA_ASSERT(ta_compare_json_with_limit(expected_json, actual_json, &options) == RA_OK, 0);
```

3.4 Compare Text While Ignoring Whitespace and Line Endings

```
const ra_text_compare_options_t options = {
    1,
    1
};
```

```
RA_ASSERT(ta_compare_text_normalized(left_text, right_text, &options) == RA_OK, 0);
```

4. Convenience Forms

- `ta_compare_paths` performs a byte-for-byte path comparison.
- `ta_compare_json` performs a strict JSON comparison using default limits.
- `ta_compare_file_lines` compares files line-by-line and returns `RA_OK` or `RA_MISMATCH`.

For the full declaration list and per-function comments, see `include/testapi.h`.